The Pinhole (Perspective) Camera

Formulas, matrices and a self-test

Based on Lecture 1, *Camera Models and Calibration* (Hajder & Chetverikov, ELTE)

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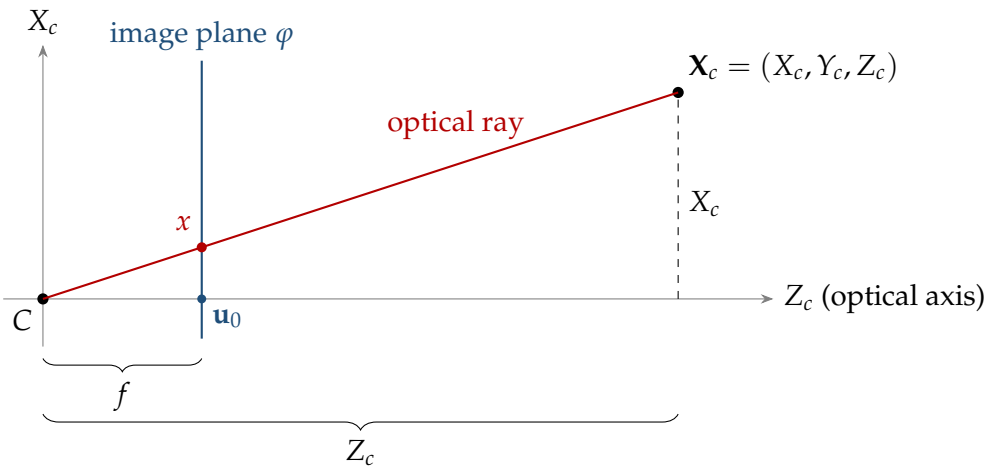
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1 Geometry and notation

Intuition

Light from every scene point passes through one single point, the **focal point** C, and hits the **image plane** at distance f. Because all rays meet at C, far-away objects look smaller: image size is proportional to $1/\text{depth}$. The whole model is "divide by depth".



C	focal point (centre of projection, camera centre)
optical ray	line joining a 3D point and C
optical axis	line through C perpendicular to the image plane φ
f	focal length: distance from C to φ
$\mathbf{u}_0 = [u_0, v_0]^T$	principal point: where the optical axis meets φ
k_u, k_v	pixel densities (pixels per unit length), horizontal / vertical

Three coordinate systems.

world	$\mathbf{X} = [X, Y, Z]^T$	homogeneous $\tilde{\mathbf{X}} = [X, Y, Z, 1]^T$
camera	$\mathbf{X}_c = [X_c, Y_c, Z_c]^T$	homogeneous $\tilde{\mathbf{X}}_c = [X_c, Y_c, Z_c, 1]^T$
image (pixels)	$\mathbf{u} = [u, v]^T$	homogeneous $\tilde{\mathbf{u}} = [u, v, 1]^T$

The pipeline is always

$$\underbrace{\mathbf{X}}_{\text{world}} \xrightarrow{R,\,\mathbf{t}} \underbrace{\mathbf{X}_c}_{\text{camera}} \xrightarrow{\text{divide by } Z_c} \text{image plane} \xrightarrow{f,k_u,k_v,u_0,v_0} \underbrace{\mathbf{u}}_{\text{pixels}}.$$

2 Homogeneous coordinates

Intuition

Perspective projection contains a *division* and the world-to-camera change contains a *translation*. Neither is a matrix multiplication in ordinary coordinates. Adding an extra coordinate turns both into plain matrix products, so the whole camera becomes one matrix.

A point $[x, y]^T$ is written as $[x, y, 1]^T$, and every non-zero multiple represents the same point:

$$[x, y, 1]^T \sim [wx,\, wy,\, w]^T \qquad (w \neq 0).$$

To go back to ordinary coordinates, **divide by the last entry**:

$$[a,\, b,\, c]^T \longmapsto \left(\tfrac{a}{c},\, \tfrac{b}{c}\right).$$

The symbol "$\sim$" means *equal up to a non-zero scale factor*. This is called the **scale ambiguity** of homogeneous coordinates.

3 Step 1: world → camera (extrinsics)

Intuition

First move the world so that the camera centre sits at the origin, then rotate it so that the camera looks along $+Z$. $\mathbf{t}$ is **where the camera is** (its centre in world coordinates); R is **how it is turned**.

World to camera

$$\mathbf{X}_c = R(\mathbf{X} - \mathbf{t}) \qquad \Longleftrightarrow \qquad \mathbf{X}_c = R\,[\,I \mid -\mathbf{t}\,] \begin{bmatrix}\mathbf{X}\\ 1\end{bmatrix} \tag{1}$$

- I is the 3×3 identity; $[\,I \mid -\mathbf{t}\,]$ is the 3×4 matrix $\begin{bmatrix}1 & 0 & 0 & -t_1\\ 0 & 1 & 0 & -t_2\\ 0 & 0 & 1 & -t_3\end{bmatrix}$.

- R is a rotation matrix: $R^T R = I$ (orthonormal) and $\det R = +1$. It has **3** degrees of freedom (e.g. three angles).
- $\mathbf{t}$ has **3** degrees of freedom. Together $R, \mathbf{t}$ are the 6 **extrinsic parameters**.
- Equivalent form often used in books/OpenCV: $\mathbf{X}_c = R\mathbf{X} + \mathbf{t}'$ with $\mathbf{t}' = -R\mathbf{t}$. Be careful which convention a problem uses!

4 Step 2: projection onto the image plane

By similar triangles in the figure, the metric image coordinate is

$$\frac{x}{f} = \frac{X_c}{Z_c} \quad \Rightarrow \quad x = f\,\frac{X_c}{Z_c}, \qquad y = f\,\frac{Y_c}{Z_c}.$$

To get pixels, multiply by the pixel density and shift by the principal point (pixel coordinates start at the image corner, not at the optical axis):

Perspective projection in pixels

$$u = \frac{fk_u}{Z_c}\,X_c + u_0, \qquad v = \frac{fk_v}{Z_c}\,Y_c + v_0 \tag{2}$$

Consequences you should be able to see directly from this formula:

- Multiplying $\mathbf{X}_c$ by any $\lambda > 0$ (sliding along the same ray) gives the *same* pixel.
- Doubling the depth Z_c (keeping X_c, Y_c) halves the offset $u - u_0$: objects look half as big.
- fk_u and fk_v are the focal length *measured in pixels*. If $k_u = k_v$ the pixels are square.

5 The intrinsic (calibration) matrix K

Intuition

K collects everything that belongs to the camera *itself* (lens + sensor) and does not change when you move the camera. It converts a direction in camera coordinates into a pixel.

Projection with homogeneous coordinates

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim K\,\mathbf{X}_c, \qquad K = \begin{bmatrix} fk_u & s & u_0 \\ 0 & fk_v & v_0 \\ 0 & 0 & 1 \end{bmatrix} \tag{3}$$

Check that it works. With $s = 0$:

$$K\mathbf{X}_c = \begin{bmatrix} fk_u X_c + u_0 Z_c \\ fk_v Y_c + v_0 Z_c \\ Z_c \end{bmatrix} \xrightarrow{\div Z_c} \begin{bmatrix} fk_u X_c/Z_c + u_0 \\ fk_v Y_c/Z_c + v_0 \\ 1 \end{bmatrix},$$

which is exactly equation (2). The homogeneous division by Z_c *is* the perspective effect.

- K is **upper triangular**.
- 5 parameters: fk_u, fk_v, u_0, v_0 and skew s.

- In practice $s = 0$ (rectangular pixels), so **4 realistic parameters**. If additionally $k_u = k_v$ (square pixels), only 3 remain.
- Changing the zoom (focal length) changes only K; moving or turning the camera changes only $R, \mathbf{t}$.

6 The full projection matrix P

Pinhole camera model

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \sim \underbrace{K\,R\,[\,I \mid -\mathbf{t}\,]}_{P\ (3\times4)} \begin{bmatrix} \mathbf{X} \\ 1 \end{bmatrix}, \qquad P = [\,KR \mid -KR\mathbf{t}\,] \tag{4}$$

- P is 3×4, so it has 12 entries, but it is only defined up to scale $\Rightarrow$ **11 degrees of freedom**: 5 in K, 3 in R, 3 in $\mathbf{t}$.
- The **left** 3×3 **block** is KR; the **last column** is $\mathbf{p}_4 = -KR\mathbf{t}$. Hence $\mathbf{t} = -R^T K^{-1}\mathbf{p}_4$ (used later in calibration).
- Row by row, with $\mathbf{p}_i^T$ the i-th row of P:

$$u = \frac{\mathbf{p}_1^T \tilde{\mathbf{X}}}{\mathbf{p}_3^T \tilde{\mathbf{X}}} = \frac{P_{11}X + P_{12}Y + P_{13}Z + P_{14}}{P_{31}X + P_{32}Y + P_{33}Z + P_{34}}, \qquad v = \frac{\mathbf{p}_2^T \tilde{\mathbf{X}}}{\mathbf{p}_3^T \tilde{\mathbf{X}}}.$$

The camera centre is the null vector of P

Plug the camera centre $\tilde{\mathbf{C}} = [\mathbf{t}^T, 1]^T$ into P:

$$P\tilde{\mathbf{C}} = KR\,[\,I \mid -\mathbf{t}\,] \begin{bmatrix} \mathbf{t} \\ 1 \end{bmatrix} = KR(\mathbf{t} - \mathbf{t}) = \mathbf{0}.$$

The vector $[0, 0, 0]^T$ is not a valid homogeneous point, so C is the **only point with no image**: every ray passes through it.

7 Back-projection: from a pixel to a ray

Intuition

Projection throws the depth away. Going backwards, a pixel gives you the whole optical ray, not one point. To fix the 3D point you need a second calibrated camera that sees the same point (a *correspondence*); intersecting the two rays is called **triangulation**.

Solve equation (2) for X_c, Y_c:

$$X_c = \frac{Z_c}{f k_u}(u - u_0), \qquad Y_c = \frac{Z_c}{f k_v}(v - v_0), \qquad Z_c = Z_c \text{ (unknown)}.$$

Back-projection in camera coordinates

$$\mathbf{X}_c = Z_c\, K^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix}, \qquad K^{-1} = \begin{bmatrix} \frac{1}{f k_u} & 0 & -\frac{u_0}{f k_u} \\ 0 & \frac{1}{f k_v} & -\frac{v_0}{f k_v} \\ 0 & 0 & 1 \end{bmatrix} \quad (s = 0) \tag{5}$$

$K^{-1}\tilde{\mathbf{u}}$ is the *direction* of the ray; letting Z_c vary traces the whole ray.

In world coordinates (homogeneous form from the lecture):

$$\tilde{\mathbf{X}}(\lambda) = (1-\lambda)\, P^{+}\tilde{\mathbf{u}} + \lambda\, \tilde{\mathbf{C}}, \qquad P^{+} = P^{T}(PP^{T})^{-1}, \quad PP^{+} = I. \tag{6}$$

The line passes through $P^{+}\tilde{\mathbf{u}}$ (at $\lambda = 0$, a point that projects onto $\mathbf{u}$) and through the camera centre $\tilde{\mathbf{C}}$ (at $\lambda = 1$, the null vector of P).

8 Worked example

Given:

$$K = \begin{bmatrix} 800 & 0 & 320 \\ 0 & 800 & 240 \\ 0 & 0 & 1 \end{bmatrix}, \qquad R = I, \qquad \mathbf{t} = [0, 0, -1]^{T} \text{ (camera centre)}, \qquad \mathbf{X} = [0.5,\ -0.25,\ 3]^{T}.$$

1. World $\rightarrow$ camera: $\mathbf{X}_c = R(\mathbf{X} - \mathbf{t}) = [0.5,\ -0.25,\ 4]^{T}$.

2. Apply K: $K\mathbf{X}_c = \begin{bmatrix} 800(0.5) + 320(4) \\ 800(-0.25) + 240(4) \\ 4 \end{bmatrix} = \begin{bmatrix} 1680 \\ 760 \\ 4 \end{bmatrix}$.

3. Divide by the last entry: $(u, v) = (1680/4,\ 760/4) = (420,\ 190)$.

Same with the full P:

$$P = KR[\, I \mid -\mathbf{t}\,] = \begin{bmatrix} 800 & 0 & 320 & 320 \\ 0 & 800 & 240 & 240 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \qquad P \begin{bmatrix} 0.5 \\ -0.25 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 1680 \\ 760 \\ 4 \end{bmatrix} \checkmark$$

Back-projection check (pretend we know $Z_c = 4$):

$$K^{-1} \begin{bmatrix} 420 \\ 190 \\ 1 \end{bmatrix} = \begin{bmatrix} (420 - 320)/800 \\ (190 - 240)/800 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.125 \\ -0.0625 \\ 1 \end{bmatrix}.$$

Multiplying by $Z_c = 4$ gives $[0.5,\ -0.25,\ 4]^{T} = \mathbf{X}_c$ ✓.

Also note $P\,[0, 0, -1, 1]^{T} = \mathbf{0}$: the camera centre is the null vector of P.

9 Summary sheet

Everything on one card

World → camera:	$\mathbf{X}_c = R(\mathbf{X} - \mathbf{t})$
Pixel coordinates:	$u = \frac{fk_u}{Z_c} X_c + u_0, \quad v = \frac{fk_v}{Z_c} Y_c + v_0$
Intrinsics:	$K = \begin{bmatrix} fk_u & s & u_0 \\ 0 & fk_v & v_0 \\ 0 & 0 & 1 \end{bmatrix}$ (5 params, 4 if $s = 0$)
Full model:	$\tilde{\mathbf{u}} \sim P\tilde{\mathbf{X}}, \quad P = KR[\,I \mid -\mathbf{t}\,] = [\,KR \mid -KR\mathbf{t}\,]$ (11 DoF)
Camera centre:	$P\tilde{\mathbf{C}} = \mathbf{0}, \quad \tilde{\mathbf{C}} = [\mathbf{t}^T, 1]^T$
Back-projection:	$\mathbf{X}_c = Z_c K^{-1}\tilde{\mathbf{u}}$ (a ray, depth unknown)
World ray:	$\tilde{\mathbf{X}}(\lambda) = (1-\lambda)P^+\tilde{\mathbf{u}} + \lambda\tilde{\mathbf{C}}, \quad P^+ = P^T(PP^T)^{-1}$

10 Self-test: multiple choice

Exactly one answer is correct for each question. Solutions with explanations are on the following pages; try all questions first.

Q1. In $P = KR[\, I \mid -\mathbf{t}\,]$, the vector $\mathbf{t}$ is

(a) the world origin expressed in camera coordinates
(b) the camera centre expressed in world coordinates
(c) the principal point in pixels
(d) the offset of the pixel origin from the image corner

Q2. How many degrees of freedom does a general pinhole projection matrix P have?

(a) 12 (b) 9 (c) 11 (d) 6

Q3. In $\tilde{\mathbf{u}} \sim K\mathbf{X}_c$, the symbol "$\sim$" means

(a) equal up to a non-zero scale factor
(b) approximately equal because of noise
(c) equal after rotation
(d) proportional only in the first coordinate

Q4. $K = \begin{bmatrix} 1000 & 0 & 640 \\ 0 & 1000 & 360 \\ 0 & 0 & 1 \end{bmatrix}$ and $\mathbf{X}_c = [0.2,\ -0.1,\ 2]^T$. The pixel is

(a) $(100, -50)$ (b) $(540, 310)$ (c) $(740, 410)$ (d) $(740, 310)$

Q5. A point $\mathbf{X}_c$ is replaced by $2\mathbf{X}_c$ (it moves along its optical ray, twice as far from C). Its pixel

(a) moves twice as far from the principal point
(b) does not change
(c) moves half as far from the principal point
(d) becomes undefined

Q6. An object's depth Z_c doubles while X_c, Y_c stay the same. The distance $u - u_0$ in the image

(a) doubles (b) stays the same (c) halves (d) is divided by four

Q7. If the skew is $s = 0$ and pixels are square ($k_u = k_v$), how many free parameters remain in K?

(a) 5 (b) 4 (c) 2 (d) 3

Q8. The principal point is

(a) the point where the optical axis meets the image plane
(b) the focal point C
(c) the top-left pixel of the image
(d) the projection of the world origin

Q9. The camera centre $\tilde{\mathbf{C}}$ (homogeneous) satisfies

(a) $P\tilde{\mathbf{C}} = [0, 0, 1]^T$ (b) $\tilde{\mathbf{C}} = P^{+}\tilde{\mathbf{u}}$ (c) $P\tilde{\mathbf{C}} = \mathbf{0}$ (d) $K\mathbf{C} = \mathbf{0}$

Q10. Back-projecting a single pixel of a calibrated camera gives

(a) a unique 3D point
(b) a plane through C
(c) nothing, since P is not invertible
(d) a ray (line) through the camera centre

Q11. With K from Q4, back-project the pixel $(740, 310)$ at depth $Z_c = 5$. Then $\mathbf{X}_c =$

(a) $[0.1, -0.05, 1]^T$ (b) $[0.5, -0.25, 5]^T$ (c) $[3.7, 1.55, 5]^T$ (d) $[500, -250, 5]^T$

Q12. A projection returns the homogeneous vector $[1200, 900, 3]^T$. The pixel is

(a) $(1200, 900)$ (b) $(3600, 2700)$ (c) $(400, 300)$ (d) $(600, 450)$

Q13. $K = \begin{bmatrix} 500 & 0 & 250 \\ 0 & 500 & 250 \\ 0 & 0 & 1 \end{bmatrix}$, $R = I$, $\mathbf{t} = [0, 0, -4]^T$, world point $\mathbf{X} = [1, 2, 0]^T$. The pixel is

(a) $(375, 500)$ (b) $(125, 250)$ (c) $(500, 1000)$ (d) $(125, 0)$

Q14. Which block form of $P = KR[\, I \mid -\mathbf{t}\,]$ is correct?

(a) $[\, K \mid -R\mathbf{t}\,]$ (b) $[\, KR \mid -KR\mathbf{t}\,]$ (c) $[\, R \mid \mathbf{t}\,]$ (d) $[\, KR \mid K\mathbf{t}\,]$

Q15. Why are homogeneous coordinates used in the camera model?

(a) They remove lens distortion
(b) They make the depth of each point recoverable from one image
(c) They reduce the number of parameters of K
(d) They turn translation and perspective division into matrix multiplications

Q16. You zoom the lens in (larger f) without moving the camera. What changes?

(a) only R (b) only $\mathbf{t}$ (c) only K (d) K, R and $\mathbf{t}$

Q17. A rotation matrix R satisfies

(a) $R^T R = I$ and $\det R = +1$
(b) $R^T R = I$ and $\det R = -1$
(c) R is upper triangular
(d) $R = R^T$

Q18. The whole scene is scaled by a factor 2 about the world origin, and the camera centre $\mathbf{t}$ is also scaled by 2 (R, K unchanged). The image

(a) becomes twice as large
(b) is identical
(c) becomes half as large
(d) is shifted by $\mathbf{t}$

Q19. What is the minimum number of calibrated cameras needed to triangulate a 3D point from image correspondences?

(a) 1 (b) 3 (c) 4 (d) 2

Q20. $K = \mathrm{diag}(100, 100, 1)$, $\mathbf{t} = \mathbf{0}$, and $R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (90° about Z). The world point $[1, 0, 5]^T$ projects to

(a) $(20, 0)$ (b) $(0, 20)$ (c) $(0, 100)$ (d) $(-20, 0)$

Q21. For $s = 0$, the third column of K^{-1} is

(a) $\left[-\frac{u_0}{fk_u},\ -\frac{v_0}{fk_v},\ 1\right]^T$

(b) $[u_0,\ v_0,\ 1]^T$

(c) $\left[\frac{u_0}{fk_u},\ \frac{v_0}{fk_v},\ 1\right]^T$

(d) $[-u_0,\ -v_0,\ 1]^T$

Q22. A straight 3D line (not through C) is imaged by a pinhole camera as

(a) a circle arc (b) a single point (c) a parabola (d) a straight line

11 Solutions

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
b	c	a	d	b	c	d	a	c	d	b
Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20	Q21	Q22
c	a	b	d	c	a	b	d	b	a	d

Q1. **(b)** From $\mathbf{X}_c = R(\mathbf{X} - \mathbf{t})$: the world point $\mathbf{X} = \mathbf{t}$ is mapped to $\mathbf{X}_c = \mathbf{0}$, the camera origin. So $\mathbf{t}$ is where the camera sits in the world.

Q2. **(c)** 11. P has 12 entries but is only defined up to scale. Equivalently 5 (in K) $+3$ (in R) $+3$ (in $\mathbf{t}$).

Q3. **(a)** Homogeneous vectors that differ by a non-zero factor represent the same point; the true pixel is obtained by dividing by the last entry.

Q4. **(d)** $u = 1000 \cdot \frac{0.2}{2} + 640 = 740$, $v = 1000 \cdot \frac{-0.1}{2} + 360 = 310$. Option (a) forgets u_0, v_0; (b) flips the sign of X_c; (c) flips the sign of Y_c.

Q5. **(b)** $u = fk_u \frac{2X_c}{2Z_c} + u_0 = fk_u \frac{X_c}{Z_c} + u_0$. All points of one optical ray land on the same pixel.

Q6. **(c)** $u - u_0 = fk_u X_c / Z_c$ is inversely proportional to Z_c: twice as far means half as big.

Q7. **(d)** 3: one focal length in pixels fk, plus u_0 and v_0.

Q8. **(a)** By definition. It is usually near, but not exactly at, the image centre; the top-left pixel is where pixel coordinates start, which is why u_0, v_0 appear.

Q9. **(c)** $P\tilde{\mathbf{C}} = KR(\mathbf{t} - \mathbf{t}) = \mathbf{0}$. C is the null vector of P.

Q10. **(d)** $\mathbf{X}_c = Z_c K^{-1}\tilde{\mathbf{u}}$ with unknown Z_c: all depths are possible, giving the optical ray through C. Depth needs a second view.

Q11. **(b)** $K^{-1}[740, 310, 1]^T = [(740 - 640)/1000,\ (310 - 360)/1000,\ 1]^T = [0.1, -0.05, 1]^T$; times $Z_c = 5$ gives $[0.5, -0.25, 5]^T$. Option (a) forgets to multiply by Z_c. (Consistent with Q4: same ray, different depth.)

Q12. **(c)** Divide by the last entry: $(1200/3,\ 900/3) = (400, 300)$.

Q13. **(a)** $\mathbf{X}_c = \mathbf{X} - \mathbf{t} = [1, 2, 4]^T$. $u = 500 \cdot \frac{1}{4} + 250 = 375$, $v = 500 \cdot \frac{2}{4} + 250 = 500$. Option (b) ignores the principal point; option (d) uses $\mathbf{X} + \mathbf{t}$ by mistake (giving $Z_c = -4$, a point behind the camera).

Q14. **(b)** $KR[\, I \mid -\mathbf{t}\,] = [\, KR\, I \mid KR(-\mathbf{t})\,] = [\, KR \mid -KR\mathbf{t}\,]$.

Q15. **(d)** Translation (+ a vector) and the perspective division by Z_c are not linear in ordinary coordinates; with one extra coordinate both become matrix products, so the whole camera is the single matrix P. Homogeneous coordinates do *not* recover depth.

Q16. **(c)** The focal length lives in K (fk_u, fk_v). R and $\mathbf{t}$ describe pose and do not change.

Q17. **(a)** Rotations are orthonormal and preserve orientation. $\det = -1$ would be a reflection; "upper triangular" describes K.

Q18. **(b)** $\mathbf{X}_c = R(2\mathbf{X} - 2\mathbf{t}) = 2R(\mathbf{X} - \mathbf{t})$, and scaling $\mathbf{X}_c$ does not change the pixel (Q5). This is why absolute scale cannot be recovered from images alone.

Q19. **(d)** Each camera gives one ray; two rays (from two different centres) intersect in the 3D point.

Q20. **(b)** $\mathbf{X}_c = R\mathbf{X} = [0 \cdot 1 - 1 \cdot 0,\ 1 \cdot 1 + 0,\ 5]^T = [0, 1, 5]^T$. $u = 100 \cdot 0/5 = 0$, $v = 100 \cdot 1/5 = 20$.

Q21. **(a)** From $X_c/Z_c = (u - u_0)/(fk_u) = \frac{1}{fk_u}u - \frac{u_0}{fk_u}$, and the same for v. See the explicit K^{-1} in Section 7.

Q22. **(d)** Lines are preserved: the line and C span a plane, and that plane cuts the image plane in a straight line. (Real lenses with radial distortion bend lines, but the ideal pinhole model does not.)